ILLINOIS
Grainger College of Engineering

# Safety-Aware Learning-Based Control




**ANY-TIME REGRET-GUARANTEED ALGORITHM FOR CONTROL OF LINEAR QUADRATIC SYSTEMS**

one thread of the Ph.D. thesis

**Jafar Abbaszadeh Chekan**

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Coordinated Science Lab (CSL), University of Illinois at Urbana-Champaign

# Learning inside the loop — and what can go wrong

Control actions play a **dual role**: they regulate the system **and** decide what data is learned from.










### Corrupted data

Stealthy deception attacks on water SCADA: the records the controller learns from are altered, and the corruption goes unnoticed.

### Uncertain safe set

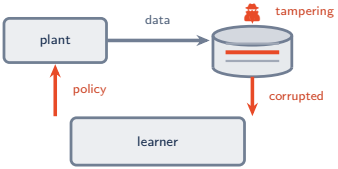
Adaptive cruise control: the admissible headway depends on a lead vehicle that must be inferred online.

### Switching under uncertainty

Personalized treatment: effects are patient-dependent, so both the choice and its timing matter.

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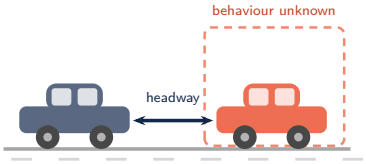
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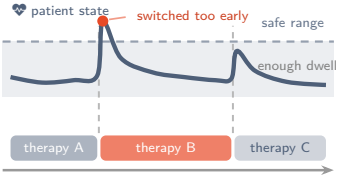
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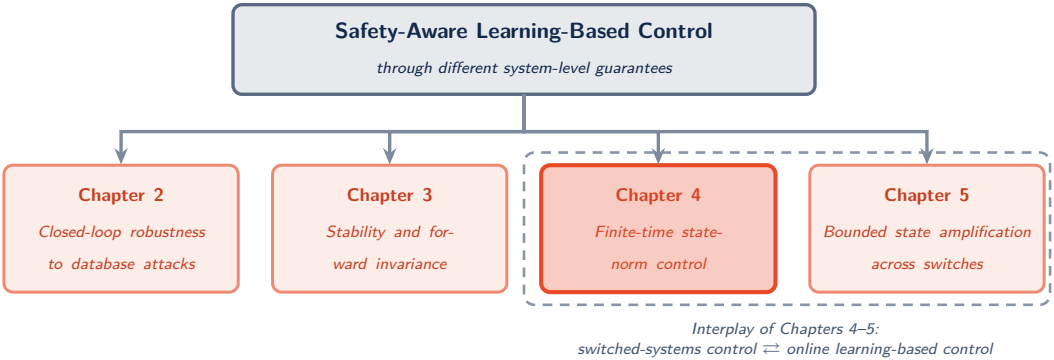
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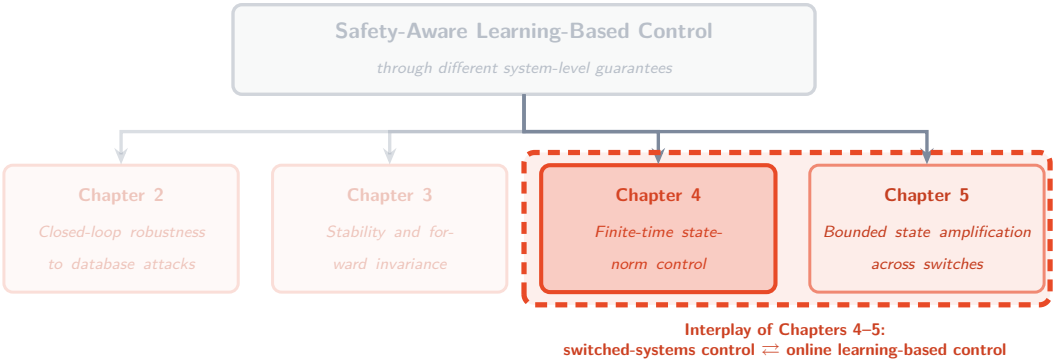
Personalized treatment: effects are patient-dependent, so both the choice and its timing matter.

# Roadmap




Each chapter develops a different system-level guarantee.

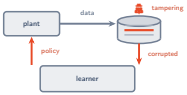
# Roadmap




The last two additionally develop the interplay between switched-systems control and online learning-based control.

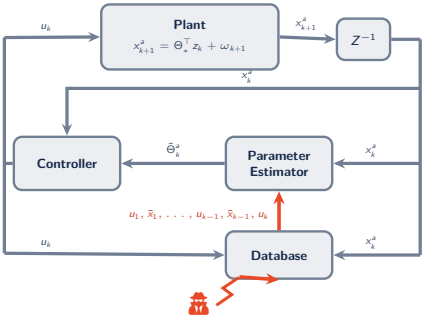
# Database attacks on the learner

COMPLETED THESIS WORK · **Chekan** & Langbort, **Automatica** 151 (2023) 110876




The adversary targets **neither sensors nor actuators**, but the database the estimator learns from.

- the confidence set built from poisoned data is no longer reliable
- how must it be **re-shaped** to absorb the injected data?
- what does the attack cost in **regret**?



# Uncertainty-dependent safe sets

COMPLETED THESIS WORK · **Chekan** & Langbort, **ACC 2023**, pp. 1264–1270

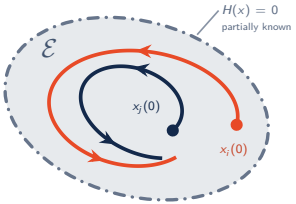



**Goal:** design a stabilizing control such that $\mathcal{E}$ is forward invariant.

**Our problem:** an **uncertainty-dependent** constraint

$$\dot{x} = \underbrace{f(x) + g(x)u}_{\text{known model}} + \underbrace{d(x)}_{\text{unknown term}}$$

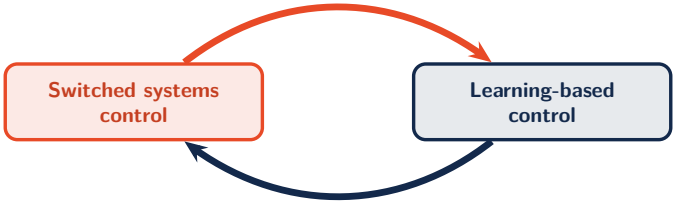
$$\mathcal{E} = \Big\{ x \in \mathbb{R}^n \;\Big|\; \underbrace{h(x) + \mathcal{F}\big(d(x)\big)}_{H(x)} \geq 0 \Big\}, \qquad \mathcal{F} \text{ is a known operator}$$




The safe set's own definition depends on the unknown term — so $H$ must be estimated and bounded with high probability from state measurements alone.

# Setting the stage

## Interaction of RL and Switched Systems Control




1. How a mindful switching strategy can be useful in learning-based control

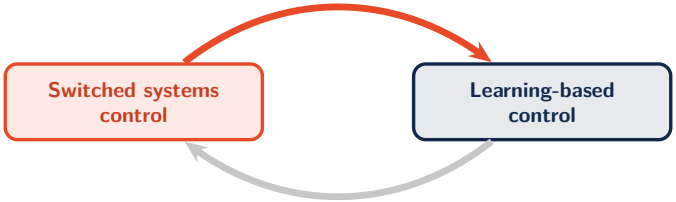
   Switched systems control $\longrightarrow$ Learning-based control

2. How online learning can can contribute to switched systems control

   Learning-based control $\longrightarrow$ Switched systems control

# Setting the stage

## The overarching goal of this presentation




1. **How a mindful switching strategy can be useful in learning-based control**

   Switched systems control $\longrightarrow$ Learning-based control

2. How online learning can can contribute to switched systems control

   Learning-based control $\longrightarrow$ Switched systems control

## Preliminaries

### Optimal LQR problem in infinite horizon

$$\begin{aligned} \text{minimize}_{\{u_t\}_{t=0}^{T-1}} \quad & \lim_{T\to\infty} \frac{1}{T}\mathbb{E}[\sum_{t=0}^{T-1} x_t^\top Q x_t + u_t^\top R u_t] \\ \text{S.t.} \quad & x_{t+1} = A_* x_t + B_* u_t + \omega_{t+1} \end{aligned}$$

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$$\text{S.t.} \quad x_{t+1} = A_* x_t + B_* u_t + \omega_{t+1}$$

**Assumptions**

- $(A_*, B_*)$ controllable
- $\omega_{t+1} \sim \mathcal{N}(0, W)$

**Solution**

- optimal sequence $u^* = \{u_t^*\}_{t=1}^{T-1}$, where $u_t^* = K_* x_t$
- $K_*$ from the solution $P_*$ of the Discrete Algebraic Riccati Equation
- $J_* = P_* \bullet W$

## Setting the stage

Regret of an arbitrary algorithm $\mathcal{A}$ that outputs $u = \{u_t\}_{t=0}^{T-1}$

$$R_{\mathcal{A}}(T) = \sum_{t=0}^{T-1} \left(x_t^\top Q x_t + u_t^\top R u_t - J_*\right).$$

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$$R_{\mathcal{A}}(T) = \sum_{t=0}^{T-1} \left(x_t^\top Q x_t + u_t^\top R u_t - J_*\right).$$

**The general goal of the literature is to design an algorithm $\mathcal{A}_*$ that minimizes $R_{\mathcal{A}}(T)$ subject to dynamic of the system:**

$$x_{t+1} = A_* x_t + B_* u_t + \omega_{t+1}$$

**when $(A_*, B_*)$ is unknown.**

# Setting the stage

$$\begin{aligned} \text{minimize}_{\{u_t\}_{t=0}^{T-1}} \quad & \sum_{t=0}^{T-1} \left( x_t^\top Q x_t + u_t^\top R u_t - J_* \right) \\ \text{S.t.} \quad & x_{t+1} = A_* x_t + B_* u_t + \omega_{t+1} \end{aligned}$$

**Assumptions**

- $A_*$ and $B_*$ are unknown
- $(A_*, B_*)$ is controllable
- $\omega_{t+1} \sim \mathcal{N}(0, W)$ and $W = \sigma_\omega^2 I$
- Access to a stabilizing policy $K_0$

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**Compact form**

$$x_{t+1} = \Theta_*^\top z_t + \omega_{t+1}, \quad z_t = \begin{pmatrix} x_t \\ u_t \end{pmatrix}$$

$$\Theta_*^\top = (A_*, \ B_*)$$

# Regret alone is not enough

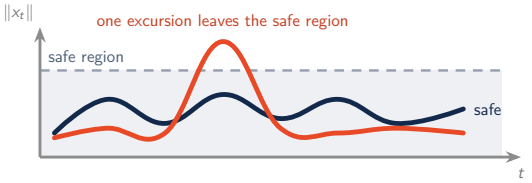
Online LQ guarantees are stated as a bound on $R(T)$ alone.

Abbasi-Yadkori & Szepesvári; Dean et al.; Cohen et al.; Mania et al.; Simchowitz & Foster.

$R(T)$ is **cumulative** — it bounds the total cost, not the trajectory that produces it.

Low regret is compatible with

- large transients while exploring,
- leaving a region it had to stay in,
- $\|x_t\|$ amplified at every update.




A **second guarantee** is needed: $\|x_t\|$ bounded w.h.p. at **every** $t$, while learning.

# The goal of this talk

Attain low regret and a controlled state norm **jointly**, **throughout** learning and without knowing the horizon $T$.

Existing SDP-based methods need a known $T$.

### The state norm needs its own mechanism

Switching amplifies $\|x_t\|$ even between stabilizing gains; a careless update blows the state up — which costs **regret** too.

Two classical routes, each falling short:

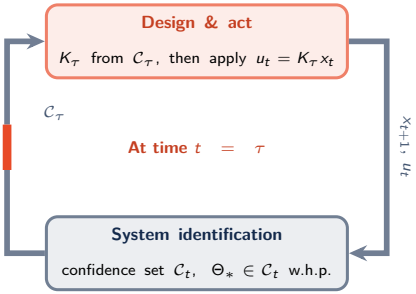
- **common Lyapunov** — no explicit $\|x_t\|$ bound
- **strong sequential stability** — a bound, but inefficient

### The questions this raises

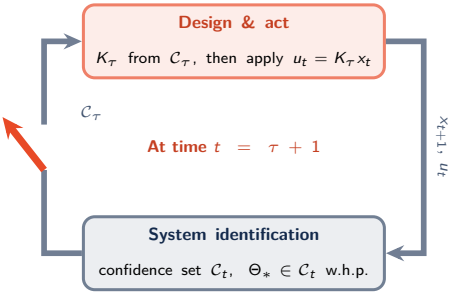
**1.** Can we do away with these notions?

**2.** Is there any trade-off between the regret bound and how tight the state-norm bound is?

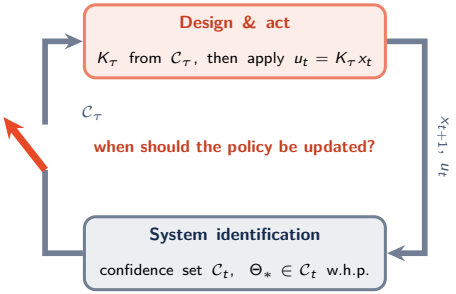
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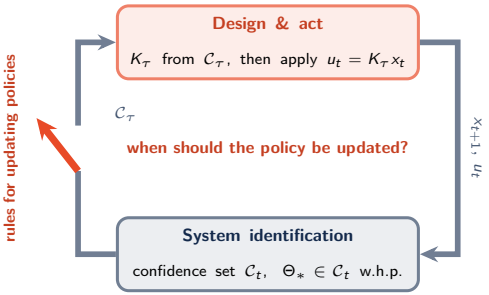
# A possible lever: how often to update




**A possible lever:** control **how often** the policy is updated.

- too fast — state explosion, bad regret scaling
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**A possible lever:** control **how often** the policy is updated.

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But *how often we may update* is itself tied to **how far apart** two consecutive policies are: the closer they sit, the sooner it is safe to switch.

So the two are **one** design problem — and both must be chosen to meet the goal.

# Outline




1. **Preliminaries — OFU-based and SDP-based algorithms**
2. Any-time regret-guaranteed algorithm — ARSLO
3. Avoiding strong sequential stability — ARSLO$^{+}(\bar{\rho})$
4. Trading regret against the state-norm bound
5. Conclusion
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## Preliminaries: OFU-based algorithm

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- Couples estimation and control design procedures
  - **Estimation** – build a confidence set for the system parameters
  - **Control design** – optimistically play with respect to that set

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First regret (non-asymptotic) guaranteed algorithm for LQ control by Abbasi and Szepesvári (2011)

Abbasi-Yadkori, Yasin, and Csaba Szepesvári. "Regret bounds for the adaptive control of linear quadratic systems." COLT, pp. 1–26, 2011.

# OFU-based algorithm by Abbasi and Szepesvári (2011)

**Control design by OFU**




**Confidence bound construction**

$$\mathcal{C}_t(\delta) = \{\Theta \in \mathbb{R}^{n\times(n+m)} \mid \mathrm{Tr}((\hat{\Theta}_t - \Theta)^\top V_t(\hat{\Theta}_t - \Theta)) \le \beta_t(\delta)\}$$

$$\beta_t(\delta) = \left(nL\sqrt{2\log\left(\frac{\det(V_t)^{1/2}}{\delta\det(\lambda I)^{1/2}}\right)} + \lambda^{1/2}\vartheta\right)^2, \quad \hat{\Theta}_t = V_t^{-1}\sum_{s=0}^{t-1} z_s x_s^\top, \quad V_t = \lambda I + \sum_{s=0}^{t-1} z_s z_s^\top$$

- $\Theta_* \in \mathcal{C}_t(\delta)$ with probability at least $1-\delta$

# OFU-based algorithm by Abbasi and Szepesvári (2011)

**Control design by OFU**

- Solve DARE for $\tilde{\Theta}_\tau$ such that

$$J(\tilde{\Theta}_\tau) \le \inf_{\Theta \in \mathcal{C}_\tau(\delta) \cap S} J(\Theta) + \frac{1}{\sqrt{\tau}}, \quad J(\Theta) = \mathrm{trace}(P(\Theta))$$

- Implement $u_t = K(\tilde{\Theta}_\tau) x_t$

$\mathcal{C}_\tau(\delta)$

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How often? $\mathcal{C}_\tau(\delta)$

$x_{t+1}$, $u_t$

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# Literature Review

Computational complexity of the OFU-based algorithm — the main motivation of the literature.

| Work | Convex | Method | Regret |
|---|---|---|---|
| Abbasi and Szepesvári (2011) | No | OFU | $\mathcal{O}(\sqrt{T})$ |
| Dean et al. (2018) | Yes | robust adaptive | $\mathcal{O}(T^{2/3})$ |
| Cohen et al. (2019) | Yes | OFU/SDP | $\mathcal{O}(\sqrt{T})^*$ |
| Mania et al. (2019) | Yes | certainty equivalence | $\mathcal{O}(\sqrt{T})$ |
| Shirani et al. (2020) | Yes | $\epsilon$-greedy | $\mathcal{O}(\sqrt{T})$ |
| Simchowitz and Foster (2020) | Yes | certainty equivalence | $\mathcal{O}(\sqrt{T})$ |
| Abeille and Lazaric (2020) | Yes | OFU | $\mathcal{O}(\sqrt{T})$ |
| Kargin et al. (2022) | Yes | Thompson sampling | $\mathcal{O}(\sqrt{T})$ |
| **Our work** | **Yes** | **OFU/SDP** | $\mathcal{O}(\sqrt{T})$ |

**O**ptimistic **S**emi-definite programming for **L**q c**O**ntrol (**OSLO**) algorithm by Cohen et al. is hindered by non-adaptivity

Cohen, Koren, and Mansour. "Learning Linear-Quadratic Regulators Efficiently with only $\sqrt{T}$ Regret." ICML, 2019.

# Preliminaries: SDP formulations of LQR

## Primal SDP formulation

$$\min_{\Sigma} \quad \begin{pmatrix} Q & 0 \\ 0 & R \end{pmatrix} \bullet \Sigma$$
$$\text{s.t.} \quad \Sigma_{xx} = \Theta_*^\top \Sigma \Theta_* + W, \quad \Sigma \succ 0$$

- $\Sigma = \left(\begin{smallmatrix} \Sigma_{xx} & \Sigma_{xu} \\ \Sigma_{ux} & \Sigma_{uu} \end{smallmatrix}\right)$, optimal solution $\Sigma^*(\Theta_*)$
- optimal feedback $K_*(\Theta_*) = \Sigma^*_{ux} \, \Sigma^{*\,-1}_{xx}$

## Dual SDP formulation

$$\max_{P} \quad P \bullet W$$
$$\text{s.t.} \quad \begin{pmatrix} Q - P & 0 \\ 0 & R \end{pmatrix} + \Theta_* P \, \Theta_*{}^\top = 0, \quad P \succeq 0$$

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- optimal average cost $J_* = \sigma_\omega^2 \operatorname{trace}\Big(P_*(\Theta_*)\Big)$

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## What Cohen et al. relax

$\Theta_*$ is unknown: both programs move to the confidence set and their equalities loosen to inequalities.

$$\text{primal:} \quad \Sigma_{xx} = \Theta_*^\top \Sigma \Theta_* + W \quad \longrightarrow \quad \Sigma_{xx} \succeq \hat{\Theta}_t^\top \Sigma \hat{\Theta}_t + W - \mu_t \Big(\Sigma \bullet V_t^{-1}\Big) I$$

$$\text{dual:} \quad \left(\begin{smallmatrix} Q-P & 0 \\ 0 & R \end{smallmatrix}\right) + \Theta_* P \Theta_*{}^\top = 0 \quad \longrightarrow \quad \left(\begin{smallmatrix} Q-P & 0 \\ 0 & R \end{smallmatrix}\right) + \hat{\Theta}_t P \hat{\Theta}_t^\top \succeq \mu_t \|P\| \, V_t^{-1}$$

# Outline




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# Outline of ARSLO algorithm

**Control design by relaxed-SDP problem**

$\mathcal{C}_\tau(\delta)$

**Confidence bound construction**

$$\mathcal{C}_t(\delta) := \left\{\Theta \in \mathbb{R}^{(n+m)\times n} \mid \mathsf{Tr}\left((\Theta - \hat{\Theta}_t)^\top V_t (\Theta - \hat{\Theta}_t)\right) \le r_t\right\}$$

$$r_t = \left(\sigma_\omega \sqrt{2n \log \frac{n \det(V_t)}{\delta \det(\ \lambda\ I)}} + \sqrt{\lambda}\ \epsilon\right)^2$$

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- $\Theta_* \in \mathcal{C}_t(\delta)$ with probability at least $1 - \delta$

# Outline of ARSLO algorithm

**Control design by relaxed-SDP problem**

$$\begin{aligned} \min \quad & \begin{pmatrix} Q & 0 \\ 0 & R \end{pmatrix} \bullet \Sigma \\ \text{s.t.} \quad & \Sigma_{xx} \succeq \hat{\Theta}_\tau^\top \Sigma \hat{\Theta}_\tau + W - \mu_\tau \left( \Sigma \bullet V_\tau^{-1} \right) I, \\ & \Sigma \succ 0 \end{aligned}$$

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- Implement $u_t = K(\mathcal{C}_\tau)x_t + \eta_t, \quad \eta_t \sim \mathcal{N}\left(0, \ \Gamma_t \ \right)$

$\mathcal{C}_\tau(\delta)$

**Confidence bound construction**

$$\mathcal{C}_t(\delta) := \left\{ \Theta \in \mathbb{R}^{(n+m)\times n} \mid \mathsf{Tr}\left( (\Theta - \hat{\Theta}_t)^\top V_t (\Theta - \hat{\Theta}_t) \right) \le r_t \right\}$$

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How often?

$\mathcal{C}_\tau(\delta)$

$x_{t+1}, \ u_t$

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$\det(V_t) \geq 2\det(V_+)$

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Specification of $\Gamma_t$, $\Theta_0$ (i.e. $\epsilon$) and $\lambda$?

$$\hat{\Theta}_t = V_t^{-1} \left( \sum_{s=1}^{t-1} z_s x_{s+1}^\top + \lambda \ \Theta_0 \ \right), \quad V_t = \ \lambda \ I + \sum_{s=0}^{t-1} z_s z_s^\top, \quad \mu_t = r_t + \sqrt{r_t}\, \vartheta \| V_t \|^{1/2}$$

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# Closed-loop Stability

**Theorem**

With the selected parameters $\lambda > 1$, $\Theta_0$ in an $\epsilon$-neighborhood of $\Theta_*$ with $\epsilon = \mathcal{O}\big(\min\big\{\frac{1}{\kappa_1^{10}}, \frac{1}{\sqrt{\lambda}}\big\}\big)$ and $\kappa_1 = \sqrt{2\|P(\mathcal{C}_1)\|/\alpha_0}$, and exploration covariance

$$\Gamma_t = \frac{\sigma_\omega^2 \bar{p}_t}{\sqrt{t+\bar{c}_t}}\Big(K(\mathcal{C}_t)K(\mathcal{C}_t)^\top + \frac{\|P(\mathcal{C}_t)\|}{\alpha_0} I\Big), \qquad \bar{p}_t = \mathcal{O}\Big(\frac{\mu_t\, \kappa_t^{10}}{\sigma_\omega^2 \sqrt{t}}\Big)$$

where $P(\mathcal{C}_t)$ is the solution of the relaxed dual SDP over $\mathcal{C}_t$.

**A**RSLO guarantees $(\kappa_*, \gamma_*)$-**strong sequential stability** of the closed-loop system, with $\kappa_* = \sqrt{2\|P(\Theta_*)\|/\alpha_0}$ and $\gamma_* = 1/(2\kappa_*^2)$, with probability at least $1-\delta$.

# Strong Sequential Stability

**Definition: $(\kappa, \gamma)$-strong sequential stability**

**strong stability** — each closed loop is a contraction in its own coordinates:

$$A_* + B_* K_t = H_t L_t H_t^{-1}, \qquad \|L_t\| \le 1 - \gamma, \quad \|K_t\| \le \kappa, \quad \|H_t\| \le B_0, \ \ \|H_t^{-1}\| \le 1/b_0, \ \ \kappa = B_0/b_0.$$

**sequentiality** — consecutive coordinate systems are close: $\|H_{t+1}^{-1} H_t\| \le 1 + \gamma/2$.

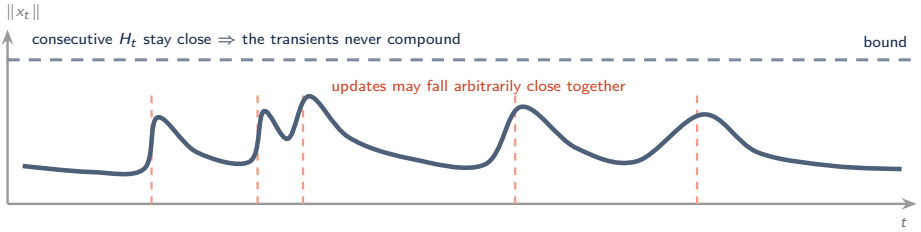
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The bound holds for **every** choice of update times.

# Further guarantees

**ARSLO tunes its parameters**

With $H_t = P^{1/2}(\mathcal{C}_t)$,

$$\mu_t \|P(\mathcal{C}_t)\| V_t^{-1} \preceq \frac{\alpha_0}{32\kappa_t^8} I$$

$$\kappa_t = \sqrt{2\|P(\mathcal{C}_t)\|/\alpha_0}$$

$\Longrightarrow$

**both halves of strong sequential stability**

$\kappa_t \le \kappa_*$ strong stability

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**Proposition**

Under ARSLO the closed-loop state norm is bounded as

$$\|x_t\| \le \kappa_* e^{-\gamma_*(t-1)/2}\|x_1\| + \frac{2\kappa_*}{\gamma_*} \max_{1\le s\le t} \|B_*\eta_s + \omega_{s+1}\|$$

with probability at least $1 - 2\delta$.

# Regret bound guarantee: ARSLO

**Theorem**

Fix $\delta \in (0,1)$. With probability at least $1-\delta$:

$$R_{\text{ARSLO}}(t) \leq \mathcal{O}\Big(\sqrt{n^2(n+m)^3\|P_*\|^{16}\ t\log^4\tfrac{t}{\delta}}\Big)$$

Terms involving $\lambda$ enter only at order $\mathcal{O}(\log\frac{t}{\delta})$, so they do not affect the $\mathcal{O}(\sqrt{t})$ leading term.

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**Convex OFU-based and certainty-equivalence alternatives**

$$R_{\text{LAG-LQ}} \leq \mathcal{O}\Big(\boldsymbol{D}^{\frac{3}{2}}\sqrt{n(n+m)^2\|P_*\|^4\ T \log^3 \tfrac{T}{\delta}}\Big) \qquad \text{Abeille and Lazaric (2020)}$$

where $\text{Tr}(P_*) \leq \boldsymbol{D}$ is an **a priori** known bound.

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$$R_{\text{CE}} \leq \mathcal{O}\Big(\sqrt{n\,m^2\|P_*\|^{11}\ T\log\tfrac{T}{\delta}}\Big) \qquad \text{Simchowitz and Foster (2020)}$$

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# Outline


1. Preliminaries — OFU-based and SDP-based algorithms
2. Any-time regret-guaranteed algorithm — ARSLO
3. **Avoiding strong sequential stability — ARSLO$^{+}(\bar{\rho})$**
4. Trading regret against the state-norm bound
5. Conclusion
6. Regret-guaranteed safe switching
7. Closing the loop

# Revisiting the Notion of Strong Sequential Stability

One threshold $\eta$ governs the policy-update rule, and it decides how close two consecutive policies must be:

$$\mu_t \|P(\mathcal{C}_t)\| V_t^{-1} \preceq \eta I \quad \Longrightarrow \quad \|P^{-1/2}(\mathcal{C}_{t+1}) P^{1/2}(\mathcal{C}_t)\| \leq 1 + \xi$$




$$\eta = \frac{\alpha_0}{32\kappa_t^8}$$

$$1 + \xi = 1 + \gamma_*/2$$

policies stay close — the policy may be updated at any time

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$\eta = \frac{\alpha_0}{4}$

$1 + \xi = 1 + 2\kappa_*^3$

policies can be far apart — an update may blow up the state; needs an extra mechanism (next)

$\eta = \frac{\alpha_0}{32\kappa_t^8}$

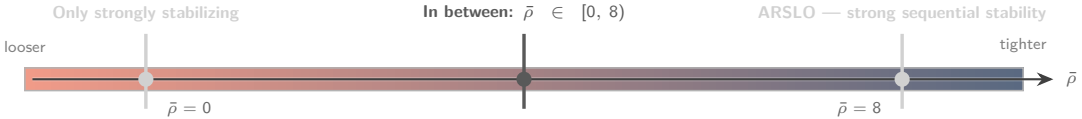
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# When a generated policy is strongly stabilizing

Tuned for a fixed $\bar{\rho}$, we call the algorithm **ARSLO$^+$($\bar{\rho}$)**.

**Theorem**

Fix $\bar{\rho} \in [0, 8)$. Suppose the warm-up returns $\Theta_0$ with $\|\Theta_0 - \Theta_*\|_F \le \underline{\epsilon}(\bar{\kappa}_1, \bar{\rho})$, and inject $\eta_t \sim \mathcal{N}(0, \Gamma_t)$, where

$$\Gamma_t = \frac{\sigma_\omega^2 \bar{p}_t}{\sqrt{t + \bar{c}_t}} \left( K(\mathcal{C}_t) K(\mathcal{C}_t)^\top + \frac{\|P(\mathcal{C}_t)\|}{\alpha_0} I \right), \qquad \bar{p}_t = \mathcal{O}\left( \frac{\mu_t \, \kappa_t^{\bar{\rho}+2}}{\sigma_\omega^2 \sqrt{t}} \right)$$

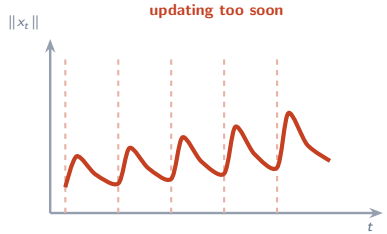
$$\lambda = \mathcal{O}\left( (n+m)^2 \, \bar{\kappa}_1^8 \log \tfrac{1}{\delta} \right)$$

with an explicit $\underline{\epsilon}(\bar{\kappa}_1, \bar{\rho})$, decreasing in $\bar{\kappa}_1$, and $\bar{\kappa}_1 = \sqrt{\kappa_1^2 + 4\kappa_1^{6-\bar{\rho}}}$.

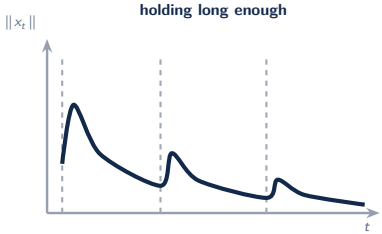
For any $\delta \in (0, 1/2)$, with probability at least $1 - 2\delta$, every policy generated by **ARSLO$^+$($\bar{\rho}$)** is $(\kappa_*, \gamma_*)$-**strongly stabilizing**, with $\kappa_* = \sqrt{2\|P_*\|/\alpha_0}$ and $\gamma_* = 1/(2\kappa_*^2)$.

# Mindful switching: a dwell-time condition

Loosening $\bar{\rho}$ lets two consecutive policies sit further apart — and switching between them does not by itself keep $\|x_t\|$ under control.



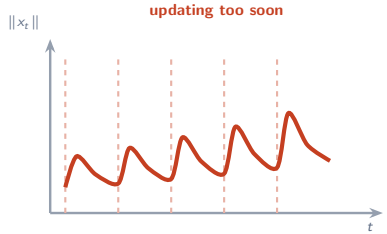

each policy is stabilizing, yet the state grows



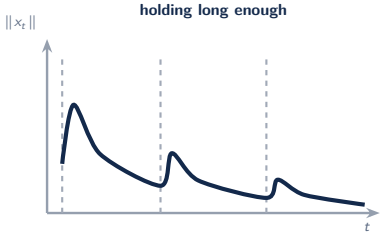

each transient decays before the next update

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each policy is stabilizing, yet the state grows




each transient decays before the next update

So we **delay** the update to let the transient settle — a **dwell time**, measured in information rather than in time:

$$\det(V_t) > \big(1 + \beta(\bar{\rho})\big)\det(V_\tau)$$

# Exploration-exploitation trade-off: a dwell-time condition

How long must we dwell? The threshold $\beta(\bar{\rho})$ answers it.

**Theorem**

With the policy update criterion $\det(V_t) > \big(1 + \beta(\bar{\rho})\big) \det(V_\tau)$ and

$$\beta(\bar{\rho}) = \begin{cases} 2\big(\kappa_1^2 + \kappa_1^{6-\bar{\rho}}\big)^{\frac{6-\bar{\rho}}{4}} & \text{if } \bar{\rho} < 6, \\ 2 & \text{if } 6 \le \bar{\rho} < 8, \end{cases} \qquad \kappa_1 = \sqrt{2\|P(\mathcal{C}_1)\|/\alpha_0}$$

ARSLO$^+(\bar{\rho})$ guarantees **closed-loop state-norm boundedness**.

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ARSLO$^+(\bar{\rho})$ guarantees **closed-loop state-norm boundedness**.

## Proposition

The closed-loop state satisfies

$$\|x_t\| \le \kappa_*(1-\gamma_*)^{\frac{t-1}{2}}\|x_1\| + \frac{2\kappa_*\big(1+\xi(\kappa_*,\bar{\rho})\big)}{\gamma_*} \max_{1\le s\le t}\|B_*\eta_s + \omega_{s+1}\|$$

with probability at least $1-\delta$, where $\xi(\kappa_*,\bar{\rho}) = 2\kappa_*^{3-\bar{\rho}/2}$ — the same gap as on the spectrum.

# Outline of ARSLO$^{+}(\bar{\rho})$ algorithm

**Control design by relaxed-SDP problem**

$$\det(V_t) \geq \left(1+\beta(\bar{\rho})\right)\det(V_\tau)$$

$\mathcal{C}_t(\delta)$

$$\begin{aligned} \min \quad & \begin{pmatrix} Q & 0 \\ 0 & R \end{pmatrix} \bullet \Sigma \\ \text{s.t.} \quad & \Sigma_{xx} \succeq \hat{\Theta}_t^\top \Sigma \hat{\Theta}_t + W - \mu_t \left(\Sigma \bullet V_t^{-1}\right) I, \\ & \Sigma \succ 0 \end{aligned}$$

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- Implement $u_t = K(\tilde{\Theta}_t)x_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \Gamma_t)$

$x_{t+1},\ u_t$

**Confidence bound construction**

$$\mathcal{C}_t(\delta) := \left\{\Theta \in \mathbb{R}^{(n+m)\times n} \mid \mathsf{Tr}\left((\Theta - \hat{\Theta}_t)^\top V_t (\Theta - \hat{\Theta}_t)\right) \leq r_t\right\}$$

$$r_t = \left(\sigma_\omega \sqrt{2n \log \tfrac{n \det(V_t)}{\delta \det(\lambda I)}} + \sqrt{\lambda}\,\underline{\epsilon}\right)^2, \qquad V_t = \lambda I + \sum_{s=0}^{t-1} z_s z_s^\top, \qquad \mu_t = r_t + \sqrt{r_t}\,\vartheta \|V_t\|^{1/2}$$

$$\hat{\Theta}_t = V_t^{-1}\left(\sum_{s=1}^{t-1} z_s x_{s+1}^\top + \lambda \Theta_0\right), \qquad \|\Theta_0 - \Theta_*\|_F \leq \underline{\epsilon}(\bar{\kappa}_1, \bar{\rho})$$

- $\Theta_* \in \mathcal{C}_t(\delta)$ with probability at least $1-\delta$

## Regret bound guarantee: ARSLO$^+(\bar{\rho})$

**Theorem**

Fix $\delta \in (0,1)$ and $\bar{\rho} \in [0, 8)$. With probability at least $1-\delta$:

$$R_{\text{ARSLO}^+(\bar{\rho})}(t) \leq \mathcal{O}\left(\sqrt{M_{\bar{\rho}}(P_*)\, \max\left\{\|P_*\|^6 \beta_*^2(\|P_*\|, \bar{\rho}),\ \|P_*\|^{8+\bar{\rho}}\right\} n^2 (n+m)^3\, t \log^4 \tfrac{t}{\delta}}\right)$$

$$M_{\bar{\rho}}(P_*) = \begin{cases} \|P_*\|^{3-\bar{\rho}/2}, & 0 \leq \bar{\rho} < 6 \\ 1, & 6 \leq \bar{\rho} < 8 \end{cases} \qquad \beta_*(\|P_*\|, \bar{\rho}) = \begin{cases} \left(\|P_*\| + \|P_*\|^{(6-\bar{\rho})/2}\right)^{\frac{6-\bar{\rho}}{4}}, & \bar{\rho} < 6 \\ 2, & 6 \leq \bar{\rho} < 8 \end{cases}$$

# Regret bound guarantee: ARSLO$^+(\bar{\rho})$

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**Corollary**

The bound is optimised at $\bar{\rho}_* = 2$:

$$R_{\text{ARSLO}^+(\bar{\rho}_*)}(t) \le \mathcal{O}\left(\sqrt{n^2(n+m)^3\,\|P_*\|^{12}\, t \log^4 \tfrac{t}{\delta}}\right)$$

compare ARSLO: $\|P_*\|^{16}$ — relaxing to $\bar{\rho}_* = 2$ improves the dependence on $\|P_*\|$.

# Back to certainty equivalence: two different axes

## 1. Regret

$$R_{\mathrm{CE}} \leq \mathcal{O}\left(\sqrt{n\, m^2 \|P_*\|^{11}\ T \log \tfrac{T}{\delta}}\right)$$ Simchowitz and Foster (2020)

$$R_{\mathrm{ARSLO}^+(\bar{\rho}_*)} \leq \mathcal{O}\left(\sqrt{n^2(n+m)^3 \|P_*\|^{12}\ t \log^4 \tfrac{t}{\delta}}\right)$$ ours, at $\bar{\rho}_* = 2$

CE is better here — both in the dimensions and in $\|P_*\|$.

# Back to certainty equivalence: two different axes

## 1. Regret

$$R_{\mathrm{CE}} \le \mathcal{O}\left(\sqrt{n\, m^2 \|P_*\|^{11}\ T \log \tfrac{T}{\delta}}\right) \qquad \text{Simchowitz and Foster (2020)}$$

$$R_{\mathrm{ARSLO}^+(\bar{\rho}_*)} \le \mathcal{O}\left(\sqrt{n^2(n+m)^3 \|P_*\|^{12}\ t \log^4 \tfrac{t}{\delta}}\right) \qquad \text{ours, at } \bar{\rho}_* = 2$$

CE is better here — both in the dimensions and in $\|P_*\|$.

## 2. Control of the state

**no explicit system-theoretic state-norm bound** — Simchowitz and Foster (2020)

$$\|x_t\| \le \kappa_*(1-\gamma_*)^{\frac{t-1}{2}}\|x_1\| + \frac{2\kappa_*\left(1+\xi(\kappa_*,\vec{\rho})\right)}{\gamma_*} \max_{1\le s\le t} \|B_*\eta_s + \omega_{s+1}\| \qquad \text{ours, for } \textbf{every } t$$

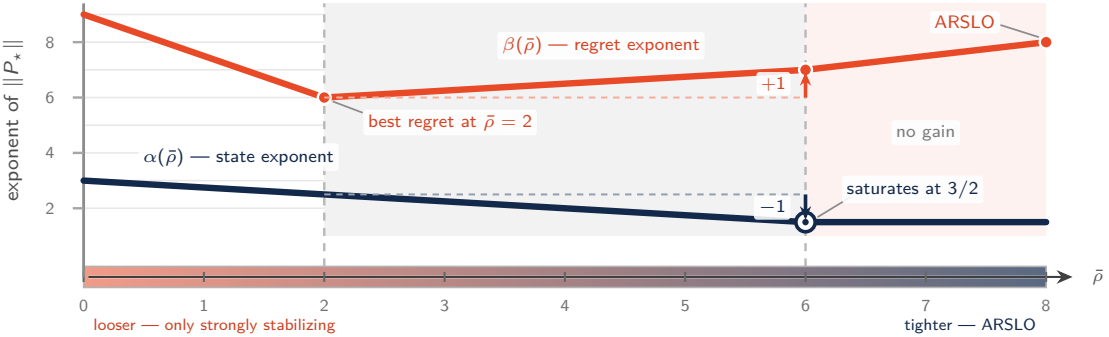
The loose bound that is available for CE depends on the initial stabilizing policy.

**The trade-off:** CE buys a better regret exponent. ARSLO$^+(\vec{\rho})$ buys an **explicit** state-norm bound, holding at **every** $t$ — which is what matters when a policy switch must not spike the trajectory.

# Outline


1. Preliminaries — OFU-based and SDP-based algorithms
2. Any-time regret-guaranteed algorithm — ARSLO
3. Avoiding strong sequential stability — ARSLO$^{+}(\vec{\rho})$
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6. Regret-guaranteed safe switching
7. Closing the loop

# State norm bound and regret bound trade-off




$$\|x_t\| \le \|P_\star\|^{\alpha(\bar\rho)} \max_{1\le s\le t} \|B_\star \eta_s + \omega_{s+1}\|$$

$$R_{\mathrm{ARSLO}^+(\bar\rho)}(t) \le \mathcal{O}\left(\|P_\star\|^{\beta(\bar\rho)} \sqrt{n^2(n+m)^3\, t\, \log^4 \tfrac{t}{\delta}}\right)$$

- regret is best at $\bar\rho = 2$ ($\beta = 6$)
- state norm improves until $\alpha$ flattens at $3/2$ ($\bar\rho \ge 6$)
- **ARSLO ($\bar\rho = 8$) buys nothing** — $\bar\rho = 6$ is already flat at $3/2$, with better regret (7 vs 8)

# Where does ARSLO stand?

| Paper | Method | Regret upper-bound w.p. $1-\delta$ | Any-time | State norm w.p. | Required information |
|---|---|---|---|---|---|
| Simchowitz & Foster (2020)$^\star$ | CE | $\widetilde{\mathcal{O}}\left(\sqrt{nm^2\|P_*\|^{11}T\log(1/\delta)}\right)$ | $\times$ | $\times$ | $K_0$ |
| Jedra & Proutiere (2022)$^\diamondsuit$ | CE | $\widetilde{\mathcal{O}}\left(\sqrt{nm^2\|P_*\|^{10.5}T\log^2 T}\right)$ | ✓ | $\times$ | $K_0$ |
| Cohen et al. (2019) | OFU | $\widetilde{\mathcal{O}}\left(\sqrt{n^4(n+m)^3D^{10}T\log^4(T/\delta)}\right)$ | $\times$ | ✓ | $K_0,\ \mathrm{Tr}(P_*)\le D$ |
| Abeille & Lazaric (2020)$^\star$ | OFU | $\widetilde{\mathcal{O}}\left(\sqrt{n(n+m)^2D^3\|P_*\|^4T\log^3(T/\delta)}\right)$ | $\times$ | $\times$ | $K_0,\ \mathrm{Tr}(P_*)\le D$ |
| **ARSLO** | OFU | $\widetilde{\mathcal{O}}\left(\sqrt{n^3(n+m)^2\|P_*\|^{16}T\log^4(T/\delta)}\right)$ | ✓ | ✓ | $K_0$ |
| **ARSLO$^+(\bar{\rho}_*)$** | OFU | $\widetilde{\mathcal{O}}\left(\sqrt{n^3(n+m)^2\|P_*\|^{12}T\log^4(T/\delta)}\right)$ | ✓ | ✓ | $K_0$ |

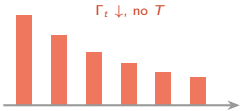
$^\diamondsuit$ This regret bound holds in expectation.

$^\star$ These works provide only a crude high-probability bound on the state norm.

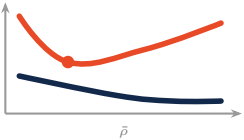
# Conclusion



Seen as a **switched system**, the update rule becomes a **design variable** — and three things follow that the literature had not identified.




**1. Any-time, by tuning the perturbation.** Driving the input-perturbation variance $\Gamma_t$ down over time removes the horizon — no $T$ need be known in advance.




**2. A genuine trade-off.** Tightening the update rule improves the state-norm bound but costs regret. The exchange is explicit in $\bar{\rho}$.



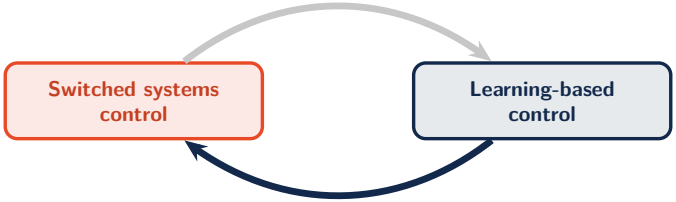

**3. Strong sequential stability is not worth its price.** A dwell-time design dominates it: $\bar{\rho} = 6$ already matches its state bound, with strictly better regret.

# Outline


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# Setting the stage

## The other direction of the interplay




1. How a mindful switching strategy can be useful in learning-based control

   Switched systems control $\longrightarrow$ Learning-based control

2. **How online learning can contribute to switched systems control**

   Learning-based control $\longrightarrow$ Switched systems control

# Regret-guaranteed safe switching

COMPLETED THESIS WORK · arXiv:2606.22223




dynamics $\quad x_{t+1} = A_*^{\sigma(t)} x_t + B_*^{\sigma(t)} u_t + \omega_{t+1},$

per-stage cost $\quad c_t^{\sigma(t)} = x_t^\top Q^{\sigma(t)} x_t + u_t^\top R^{\sigma(t)} u_t,$

$\sigma : \{0\} \cup \mathbb{N} \to \mathcal{M}$ — switching signal specifying the active mode at time $t$; $(A_\star^i, B_\star^i)$ **unknown** for every $i \in \mathcal{M}$.

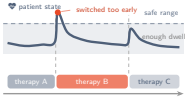



Choose the **switching times** and the **per-mode gains** so that

- $(\alpha, \beta)$-controllability: $\mathbb{E}\left[\|x_{t_{k+1}}\|^2\right] \le \alpha\, \mathbb{E}\left[\|x_{t_k}\|^2\right] + \beta\sigma_\omega^2$
- the **accumulated cost** $\mathcal{C}_{\mathcal{I}}^{\bar{M}\bar{S}} = \sum_{k=0}^{n_s} \sum_{j=t_k}^{t_{k+1}-1} c_j^{i_k}$ is minimized, $\mathcal{I} = \{i_0, \ldots, i_{n_s}\}$

# Regret-guaranteed safe switching

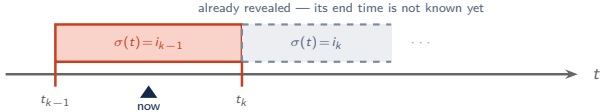
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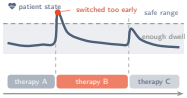



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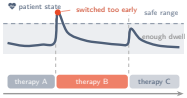



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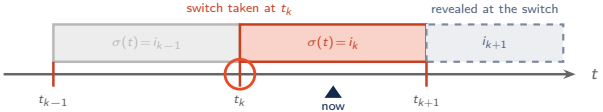
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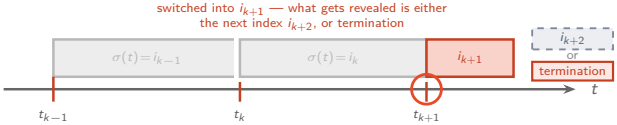
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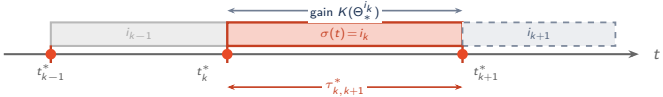
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# Regret-guaranteed safe switching




## Known modes

The dwell is available in closed form:

$$t_{k+1}^* = t_k^* + \tau_{k,k+1}^*,$$

$$\tau_{k,k+1}^* = \max\left\{1,\ -\frac{\ln \rho_*^{i_k} + \ln \rho_*^{i_{k+1}} - \ln \bar{\alpha}}{\ln\left(1-\eta_*^{i_k}\right)}\right\}$$

for every mode $i$: $P(\Theta_*^i)$ solves the DARE for that mode and the feedback $K(\Theta_*^i)$ is the gain that DARE solution yields, $\rho_*^i = \overline{\lambda}\left(P(\Theta_*^i)\right) \big/ \underline{\lambda}\left(P(\Theta_*^i)\right)$, $\eta_*^i = \underline{\lambda}\left(H(\Theta_*^i)\right) \big/ \overline{\lambda}\left(P(\Theta_*^i)\right)$, $H(\Theta_*^i) = Q^i + K^\top(\Theta_*^i) R^i K(\Theta_*^i)$.

$\Theta_*^i$ unknown

$\tau_{k,k+1}^*$, $K(\Theta_*^{i_k})$ learned from data

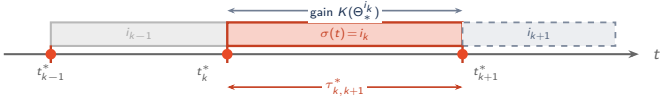
## Theorem

Expected regret against the known-mode benchmark:

$$\mathbb{E}\left[\mathcal{R}_*(\mathcal{I})\right] \lesssim \mathcal{O}\Big(n_m \log \tfrac{n_s}{\delta} + |\mathcal{M}| \log^5 \tfrac{n_s}{\delta} + |\mathcal{M}|^{1/4} n_s^{3/4} \log^2 \tfrac{n_s}{\delta}\Big)$$

$n_m$ = number of **malignant** switches: $P(\Theta_*^{i_k}) \preceq P(\Theta_*^{i_{k+1}})$.

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# Closing the loop

Safety is a **system-level guarantee** that must hold **inside the loop** while it learns.

**Chapter 2**
*Closed-loop robustness to database attacks*

**Defending the learner against data poisoning.** Regret stays guaranteed, degradation quantified.
**Chekan** & Langbort, **Automatica** 151 (2023)

**Chapter 3**
*Stability and forward invariance*

**Enforcing safety when the safe set is unknown.** The constraint is learned along the way.
**Chekan** & Langbort, **ACC** 2023

**Chapter 4**
*Finite-time state-norm control*

**Regret and the state norm trade off.** $\bar{\rho}$ is the exchange rate.
**Chekan** & Langbort, under revision, **JMLR**

**Chapter 5**
*Bounded state amplification across switches*

**Keeping the state norm under control across switches.** A learned dwell time, regret still guaranteed.
**Chekan**, Etesami & Langbort, under review, **SICON**

**Also published, outside the thesis:**

**Chekan** & Langbort, "Learn and Control While Switching: Guaranteed Stability and Sublinear Regret," **IEEE TAC** 69(12), 2024

**Chekan** et al., "Joint Stabilization and Regret Minimization through Switching in Over-Actuated Systems," **IFAC-PapersOnLine** 55(25), 2022

# Thank You!

سپاس · شكرًا · 谢谢 · धन्यवाद · Merci

Jafar Abbaszadeh Chekan · Coordinated Science Laboratory · University of Illinois Urbana-Champaign

Background: the Maragheh observatory, founded 1259 by Nasır al-Dın al-Tūsı (artist's impression)

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